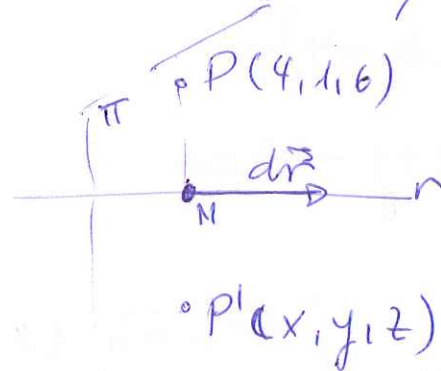


Geometria (Zuzenak eta Planuak espazioran)

10- $P(4, 1, 6)$

$$r: \begin{cases} x - y - 4z + 12 = 0 \\ 2x + y - 2z + 3 = 0 \end{cases}$$



$$\vec{dr} = \vec{n} \wedge \vec{n}' = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & -4 \\ 2 & 1 & -2 \end{vmatrix} = (6, -6, 3)$$

$$\pi: P(4, 1, 6)$$

$$\vec{n}_{\pi} = \vec{dr} = (6, -6, 3) \Rightarrow 6x - 6y + 3z + d = 0$$

$$6 \cdot 4 - 6 \cdot 1 + 3 \cdot 6 + d = 0 \Rightarrow d = -36$$

$\pi: 6x - 6y + 3z - 36 = 0$; M puntua, r eta π -ren elakete puntua da, beraz.

$$M \text{ puntua } \begin{cases} 6x - 6y + 3z - 36 = 0 \\ x - y - 4z + 12 = 0 \\ 2x + y - 2z + 3 = 0 \end{cases} \quad M = (3, -1, 4)$$

(Egater)

M P eta P' -ren edignueko dauka. $\Rightarrow$

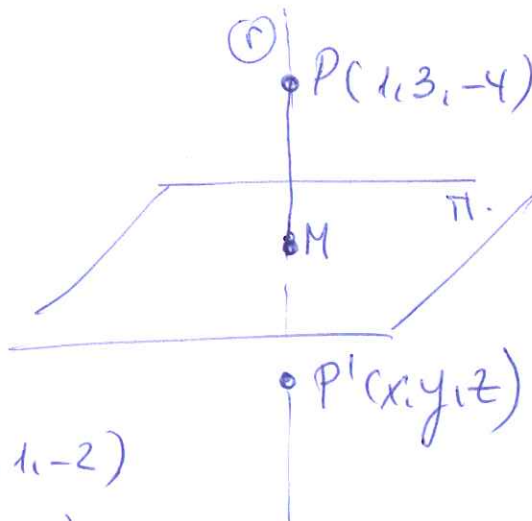
$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right)$$

$$(3, -1, 4) = \left(\frac{4+x}{2}, \frac{1+y}{2}, \frac{6+z}{2} \right) \Rightarrow \begin{cases} x = 2 \\ y = -3 \\ z = 2 \end{cases} \quad \underline{\underline{P' = (2, -3, 2)}}$$

2.0-

$$P(1, 3, -4)$$

$$\pi: 3x + y - 2z = 0$$



$$\text{rezuma: } \begin{cases} d\vec{r} = \vec{n}_\pi = (3, 1, -2) \\ R = P = (1, 3, -4) \end{cases}$$

$$r: \begin{cases} x = 1 + 3\lambda \\ y = 3 + \lambda \\ z = -4 - 2\lambda \end{cases}$$

M punctua retea π -rei e baketa punctua da, beraz.

$$M: \begin{cases} 3x + y - 2z = 0 \\ x = 1 + 3\lambda \\ y = 3 + \lambda \\ z = -4 - 2\lambda \end{cases} \Rightarrow 3(1 + 3\lambda) + (3 + \lambda) - 2(-4 - 2\lambda) = 0$$
$$\Rightarrow \lambda = -1 \Rightarrow M = (-2, 2, -2)$$

M punctua, P eta P' -rei erdizumea da, beraz.

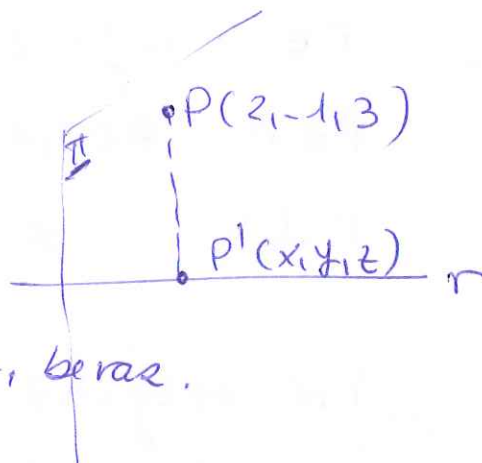
$$M = (-2, 2, -2) = \left(\frac{1+x}{2}, \frac{3+y}{2}, \frac{-4+z}{2} \right) \Rightarrow \begin{aligned} x &= -5 \\ y &= 1 \\ z &= 0 \end{aligned}$$

$$\underline{\underline{P' = (-5, 1, 0)}}$$

3.-

$$r: \begin{cases} x=3t \\ y=5t-7 \\ z=2t+2 \end{cases}$$

$$P=(2,-1,3)$$



π plano e zusemakikiko $\perp$ da, beraz.

$$\pi: \begin{cases} \vec{n}_{\pi} = d\vec{r} = (3, 5, 2) \\ P(2, -1, 3) \end{cases}$$

$$\pi: 3x + 5y + 2z + d = 0; \quad 3 \cdot 2 + 5(-1) + 2 \cdot 3 + d = 0;$$

$$d = -7$$

$$\pi: 3x + 5y + 2z - 7 = 0$$

P' puntue π eta r -ren arteko ebakete puntue da, beraz.

$$P' = \begin{cases} 3x + 5y + 2z - 7 = 0 \\ x = 3t \\ y = 5t - 7 \\ z = 2t + 2 \end{cases}$$

$$\Rightarrow 3(3t) + 5(5t - 7) + 2(2t + 2) - 7 = 0$$

$$t = 1.$$

$$\text{Beraz } \underline{\underline{P'(3, -2, 4)}}$$

4.-

$$r \in x+2y-3z-1=0$$

$$P \in r \quad P(2,1,1)$$

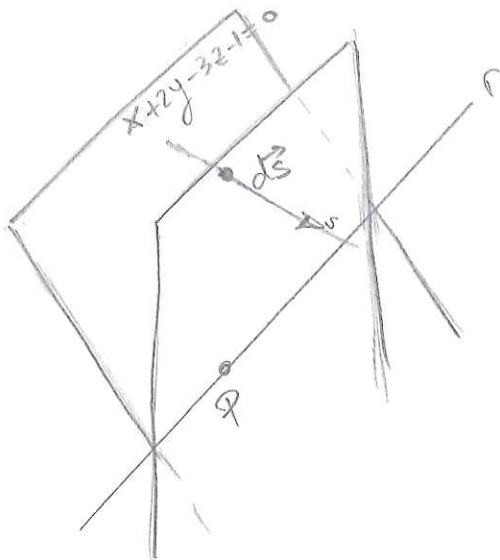
$$r \perp s \quad s: \frac{x+3}{2} = y-4 = z$$

$$r: \left\{ \begin{array}{l} \alpha: x+2y-3z-1=0 \\ \beta: \left\{ \begin{array}{l} P(2,1,1) \\ \vec{n}_\beta = d\vec{s} = (2,1,1) \end{array} \right\} \end{array} \right\} \text{ benar,}$$

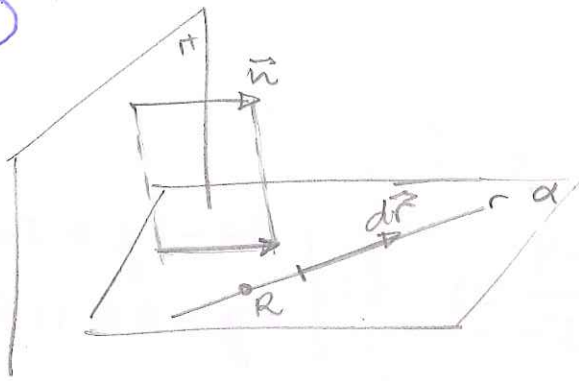
$\beta: 2x+y+z+d=0$ etc $P(2,1,1)$ puntutik pasatur, $\beta: 2 \cdot 2 + 1 \cdot 1 + 1 \cdot 1 + d = 0 \rightarrow d = -6$

$$\beta: 2x+y+z-6=0 \text{ etc } r \text{ susune.} \left\{ \begin{array}{l} x+2y-3z-1=0 \\ 2x+y+z-6=0 \end{array} \right.$$

izanup do.



5.-



$$r: \frac{x}{2} = \frac{y-1}{-1} = \frac{z+1}{3}$$

$$r: \begin{cases} R(0, 1, -1) \\ d\vec{r} = (2, -1, 3) \end{cases}$$

$$\pi: x+2y-3z+3=0 \quad \vec{n}_{\pi} = (1, 2, -3)$$

$$\alpha \text{ plawowemu osafajiek (rozupamrak)}: \begin{cases} R(0, 1, -1) \\ d\vec{r} = (2, -1, 3) \\ \vec{n}_{\pi} = (1, 2, -3) \end{cases}$$

α : równanie

$$\begin{vmatrix} x & 2 & 1 \\ y-1 & -1 & 2 \\ z+1 & 3 & -3 \end{vmatrix} = 0$$

$$-3x+9y+5z-4=0 \text{ edo}$$

$$\alpha: \underline{\underline{3x-9y-5z+4=0}}$$

$$\textcircled{60-} \quad r: \begin{cases} x+2y+z-m=0 \\ 2x-y-z+2=0 \end{cases}$$

$$l: \frac{x-1}{2} = \frac{y+1}{3} = \frac{z-4}{5} \Rightarrow l: \begin{cases} 3x-2y-5=0 \\ 5y-3z+14=0 \end{cases}$$

$$\begin{cases} r: \begin{cases} x+2y+z-m=0 \\ 2x-y-z+2=0 \end{cases} \\ s: \begin{cases} 3x-2y-5=0 \\ 5y-3z+14=0 \end{cases} \end{cases}$$

r ete s zusemak ebakitzaileak izan daitezak.

$$(S/S') = \left(\begin{array}{ccc|c} 1 & 2 & 1 & m \\ 2 & -1 & -1 & -2 \\ 3 & -2 & 0 & 5 \\ 0 & 5 & -3 & -14 \end{array} \right) \quad h(S)=h(S')=3$$

izan behar da,
beraz.

$$\begin{vmatrix} 1 & 2 & 1 & m \\ 2 & -1 & -1 & -2 \\ 3 & -2 & 0 & 5 \\ 0 & 5 & -3 & -14 \end{vmatrix} = 0 \Rightarrow m = \frac{25}{4} \quad \text{ete.}$$

$$\therefore \begin{cases} x+2y+z = 25/4 \\ 2x-y-z = -2 \\ 3x-2y = 5 \end{cases}$$

$$x = \frac{\begin{vmatrix} 25/4 & 2 & 1 \\ -2 & -1 & -1 \\ 5 & -2 & 0 \end{vmatrix}}{\begin{vmatrix} 1 & 2 & 1 \\ 2 & -1 & -1 \\ 3 & -2 & 0 \end{vmatrix}} = \frac{3}{2}$$

$$y = -1/4 \quad z = 21/4$$

$$M = \left(\frac{3}{2}, -\frac{1}{4}, \frac{21}{4} \right)$$

7.-

$$\vec{M_1 M_2} = (1, 8, 2)$$

$$\vec{M_1 M_3} = (2, 16, 4)$$

$$\vec{M_2 M_3} = (1, 8, 2)$$

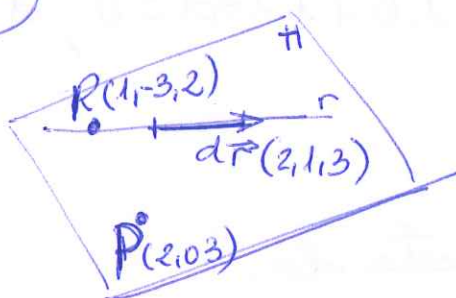
$$M_1 = (1, -8, -3)$$

$$M_2 = (2, 0, -1)$$

$$M_3 = (3, 8, 1)$$

M_1, M_2 etc M_3 punctuale alinatele deosei baze
es dade plane bat osotzen, zuzena berizik.

8.-



$$\begin{cases} r \in \Pi \\ P \in \Pi \end{cases}$$

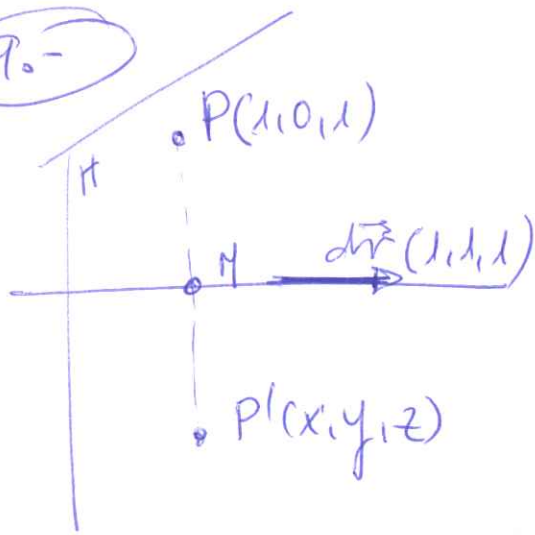
$$\Pi: \begin{cases} P = (2, 0, 3) \\ \vec{u} = \vec{dP} = (2, 1, 3) \\ \vec{v} = \vec{RP} = (1, 3, 1) \end{cases}$$

$$\Pi: \begin{vmatrix} x-2 & 2 & 1 \\ y & 1 & 3 \\ z-3 & 3 & 1 \end{vmatrix} = 0$$

$$\underline{\underline{-8x + y + 5z + 1 = 0}}$$

9.-

9.-



$$r: x-3=y=z-2$$

1) π planoua kalkulatu

$$\pi: \begin{cases} P(1, 0, 1) \\ \vec{n}_{\pi} = d\vec{r} = (1, 1, 1) \end{cases}$$

$$\pi: x+y+z+d=0 ; 1.1+1.0+1.1+d=0 ; d=-2$$

$$\pi: x+y+z-2=0$$

2) M punctue π ete r -ren ebakete de.

$$\begin{cases} x+y+z-2=0 \\ x=3+\lambda \\ y=\lambda \\ z=2+\lambda \end{cases}$$

$$\Rightarrow \lambda = -1 \Rightarrow M = (2, -1, 1)$$

3) M punctue P ete P' -ren endiguire de, beraz.

$$M = (2, -1, 1) = \left(\frac{1+x}{2}, \frac{0+y}{2}, \frac{1+z}{2} \right) \Rightarrow$$

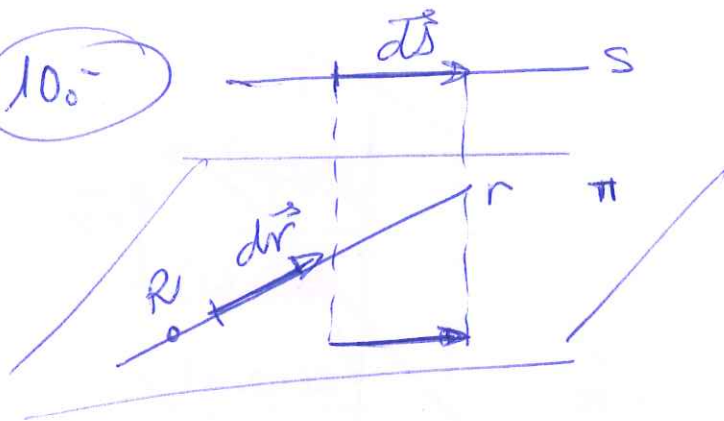
$$x=3$$

$$y=-2$$

$$z=1$$

$$\underline{\underline{P' = (3, -2, 1)}}$$

10.



π planaren ezafermak

$$\pi: \begin{cases} R \text{ puntue} \\ \vec{dr} \\ \vec{ds} \end{cases}$$

iraup dir.

a) $\vec{dr}$ lortzeko eta R puntue.

$$r: \begin{cases} x+y-2z+1=0 \\ 2x-y+z-1=0 \end{cases}$$

$$z=\lambda$$

$$x=\frac{\lambda}{3}$$

$$y=-1+\frac{5\lambda}{3}$$

$$\Rightarrow r: \begin{cases} x=\frac{\lambda}{3} \\ y=-1+\frac{5\lambda}{3} \\ z=\lambda \end{cases}$$

$$r: \begin{cases} R(0, -1, 0) \end{cases}$$

$$\vec{dr} = \left(\frac{1}{3}, \frac{5}{3}, 1\right) \text{ edo } (1, 5, 3)$$

b) $\vec{ds}$ lortzeko

$$s: \begin{cases} 2x+y-z-1=0 \\ x-y-2z+1=0 \end{cases} \quad \vec{ds} = \vec{n} \wedge \vec{n}' = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 1 & -1 & -2 \end{vmatrix} = (-3, 3, -3)$$

Beraz π planaren ekuazioa

$$\begin{vmatrix} x & 1 & -3 \\ y+1 & 5 & 3 \\ z & 3 & -3 \end{vmatrix} = 0$$

$$-24x - 6y + 18z - 6 = 0$$

$$-4x - y + 3z - 1 = 0$$

edo

$$4x + y - 3z + 1 = 0$$

110- $M(0,1,2)$

$$\alpha: \begin{cases} \vec{n}_\alpha = d\vec{r} = (2,1,0) \\ M(0,1,2) \end{cases}$$

$$\alpha: 2x + y + 0z + d = 0$$

$$2 \cdot 0 + 1 + 0 \cdot 2 + d = 0$$

$$d = -1$$

$$\alpha: \underline{2x + y - 1 = 0}$$

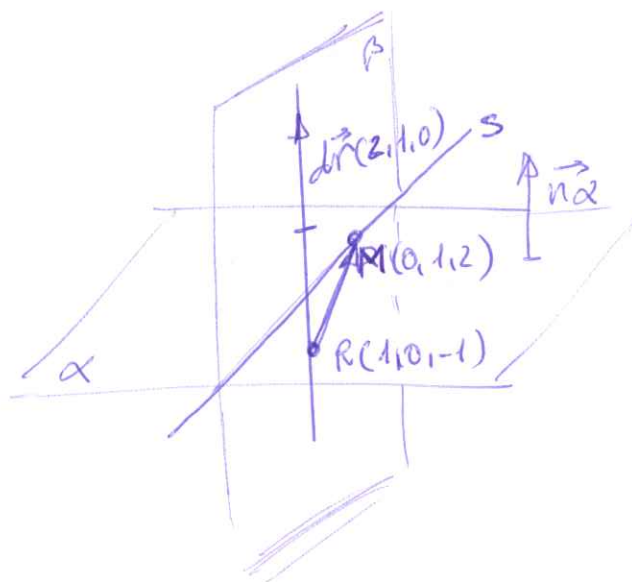
$$\beta: \begin{cases} M(0,1,2) \\ d\vec{r} = \vec{u} = (2,1,0) \\ \vec{RP} = (-1,1,3) \end{cases}$$

$$\beta: \begin{vmatrix} x-0 & 2 & -1 \\ y-1 & 1 & 1 \\ z-2 & 0 & 3 \end{vmatrix} = 0$$

$$3x - 6y + 3z - 6 + 6 = 0$$

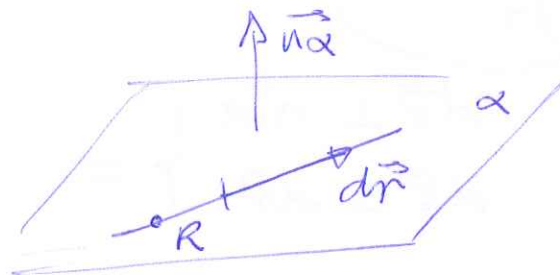
$$\beta: \underline{x - 2y + z = 0}$$

$$S \text{ line: } \begin{cases} 2x + y - 1 = 0 \\ x - 2y + z = 0 \end{cases}$$



125-

a) $r \in \alpha \Rightarrow d\vec{r} \cdot \vec{n}_\alpha = 0$



b) $R \in r$ et $R \in \alpha$.

beras.

$$d\vec{r} \cdot \vec{n}_\alpha = 0 \Rightarrow (4, -4, 1) \cdot (A, 2, -4) = 0 \Rightarrow$$

$$4A - 8 - 4 = 0 \quad \underline{\underline{A = 3}}$$

et $R(3, 1, -3) \in \alpha \Rightarrow 3 \cdot 3 + 2 \cdot 1 - 4(-3) + D = 0$

$$9 + 2 + 12 = -D$$

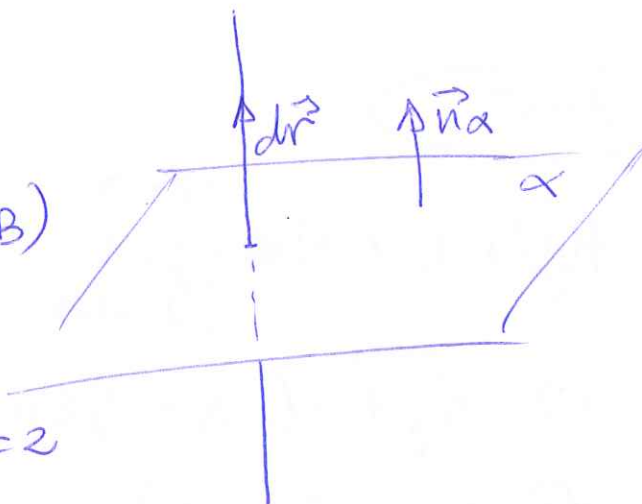
$$\underline{\underline{D = 23.}}$$

13.-

$r \perp \alpha \Rightarrow d\vec{r} = k \vec{n}_\alpha$

beras. $(A, 4, -3) = k(3, 2, B)$

$$\begin{cases} A = 3k \\ 4 = 2k \rightarrow k = 2 \\ -3 = Bk \end{cases}$$



$$\underline{\underline{A = 3 \cdot 2 = 6}}$$

$$\underline{\underline{B = \frac{k}{-3} = -\frac{2}{3}}}$$

14.-

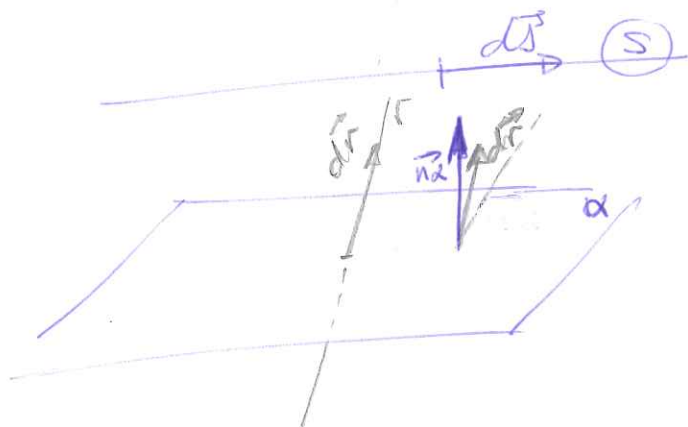
$$\begin{cases} d\vec{s} \perp \vec{n}_\alpha \\ d\vec{s} \perp d\vec{r} \end{cases} \Rightarrow$$

$$d\vec{s} = d\vec{r} \wedge \vec{n}_\alpha =$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 2 & -2 \\ 2 & -1 & 1 \end{vmatrix} = (0, -3, -3) \text{ , benar}$$

$$S: \begin{cases} M(0,1,0) \\ d\vec{s} = (0, -3, -3) \end{cases} \Rightarrow \frac{x}{0} = \frac{y-1}{-3} = \frac{z}{-3} \text{ ebo}$$

$$\frac{x}{0} = \frac{y-1}{1} = \frac{z}{1}$$

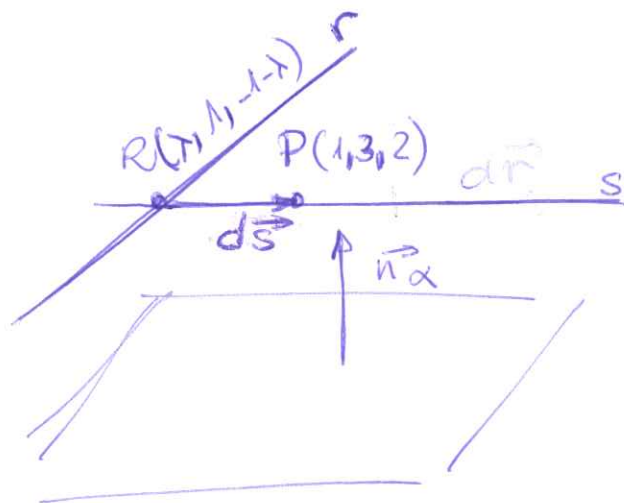


15.-

$$Rg(\lambda, 1, -1-\lambda) \Leftarrow \begin{cases} x = \lambda \\ y = 1 \\ z = -1-\lambda \end{cases}$$

$$d\vec{s} = Rg \vec{P} = (1-\lambda, 3-1, 2+1+\lambda)$$

$$d\vec{s} = (1-\lambda, 2, 3+\lambda)$$



$$d\vec{s} \perp \vec{n}_\alpha \Rightarrow d\vec{s} \cdot \vec{n}_\alpha = 0 \Rightarrow (1-\lambda, 2, 3+\lambda) \cdot (2, -1, 1) = 0$$

$$2 - 2\lambda - 1 + 3 + \lambda = 0 \rightarrow -\lambda + 3 = 0 \quad \underline{\underline{\lambda = 3}} \text{ benar}$$

$$R = D \begin{cases} x = \lambda \\ y = 1 \\ z = -1-\lambda \end{cases} \Rightarrow \underline{\underline{R = (3, 1, -4)}}$$

16.-

$$A: \begin{cases} R(3, 3, -a) \\ d\vec{r}(2, -1, 2) \end{cases}$$

$$S: \begin{cases} S(1, -1, -4) \\ d\vec{S} = (4, 3, 5) \end{cases}$$

$$(M | M') = \left(\begin{array}{cc|c} 2 & 1 & 1 \\ -1 & -1 & 0 \\ 2 & -4 & 5+a \end{array} \right)$$

$$h(M) = 2 \quad ; \quad \begin{vmatrix} 2 & 1 \\ -1 & -1 \end{vmatrix} \neq 0 \text{ daktako.}$$

Elkarkeonak izan daitezau r eta s zuzenak.
 $h(M') = 2$ izan behar da, beraz.

$$\begin{vmatrix} 2 & 1 & 1 \\ -1 & -1 & 0 \\ 2 & -4 & 5+a \end{vmatrix} = 0 \text{ izan behar duen.}$$

$$-2(5+a) + 4 + 2 + (5+a) = 0 \Rightarrow \underline{\underline{a=1}}$$

Bi zuzenak ezin dira zuzen bera izan $d\vec{r}$ eta $d\vec{S}$, zuzenen normalide bektoreak, desberdinak diralak.

$$\textcircled{16.} \quad r \perp s \Rightarrow d\vec{r} \cdot d\vec{s} = 0 \quad \text{beweis,}$$

$$\left. \begin{array}{l} d\vec{r} = (2, -1, 2) \\ d\vec{s} = (4, 3, 5) \end{array} \right\} \Rightarrow d\vec{r} \cdot d\vec{s} = (2, -1, 2) \cdot (4, 3, 5) = 0$$

$$8 - 3 + 10 \neq 0$$